

CodeGraph Status

CodeGraph — Status

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Field	Value
Revision	2
Created	2026-05-28
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Status	active
Status summary	Append-only ledger of every CodeGraph-related event for HelixCode (config changes, index regenerations, sync runs, validation probes). Per §11.4.78 (CodeGraph parent), §11.4.79 (own-org submodule inclusion), §11.4.80 (regular-update + sync automation), §11.4.45 / §11.4.56 (Status-doc shape). The weekly update + sync automation is INHERITED BY REFERENCE from the constitution submodule — invoke <code>constitution/scripts/codegraph_update.sh</code> and <code>constitution/scripts/codegraph_sync.sh</code> (never copied). Latest: 2026-07-07 Phase-4 reindex on codegraph 1.2.0 — sync GREEN, own-org symbol resolution PROVEN (MCP+CLI); third-party stale-entry purge (<code>cli_agents</code>) blocked on host process saturation.
Issues	HXC-041 (third-party <code>cli_agents</code> stale-index entries not purged — full codegraph index blocked on host process saturation)
Issues summary	HXC-041: live index still holds 36,089 <code>cli_agents</code> + 519 <code>cli_agents_resources</code> + 9 <code>github_pages_website</code> third-party files (<code>config.json</code> exclude is INERT in codegraph 1.2.0 — exclusion is <code>.gitignore</code>-driven); purge needs a from-scratch codegraph index which fork-failed on

Field	Value
	host at 4069/4096 user processes (§11.4.174 non-ours workloads). §11.4.10 credentials CLEAN (0 real .env/.pem/.key indexed).
Fixed	HXC-017 (own-org submodule inclusion in index)
Continuation	sibling Status_Summary.md carries the operator-readable digest per §11.4.56.

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Cadence + automation (§11.4.80)

Per §11.4.80, HelixCode MUST run the CodeGraph update + sync automation at least weekly (cadence floor per §11.4.45 status-digest cadence). The two canonical scripts are **inherited by reference** from the constitution submodule (per §3 submodule inheritance) and MUST be invoked at their constitution-submodule paths — never copied into HelixCode:

- `constitution/scripts/codegraph_update.sh` — `npm`-installs the latest `@colbymchenry/codegraph`, verifies `codegraph --version` reflects the new version (anti-bluff: `npm exit 0` is not proof of a working binary), and appends old/new version to this ledger.
- `constitution/scripts/codegraph_sync.sh` — after a successful update runs `codegraph status → codegraph sync . → codegraph status → validation`, appending every step's output to this ledger.

Regeneration mechanism (per §11.4.77): `.codegraph/codegraph.db` is gitignored; `codegraph index .` (full) or `codegraph sync .` (incremental) regenerates it from `.codegraph/config.json` (tracked).

Index configuration (§11.4.79)

`.codegraph/config.json` (tracked) controls which paths enter the index. Per §11.4.79 the `dependencies/` tree is split by submodule ownership:

INCLUDED — own-org submodules (full CLI access via vasic-digital + HelixDevelopment):

- dependencies/vasic-digital/** — ~55 own-org submodules (EventBus, Concurrency, Observability, Auth, Storage, VectorDB, Embeddings, Database, Cache, Messaging, Formatters, MCP_Module, RAG, Memory, Optimization, Plugins, Agentic, LLMops, SelfImprove, Planning, Benchmark, ToolSchema, SkillRegistry, Models, LLMProvider, BackgroundTasks, DocProcessor, conversation, LLMOrchestrator, VisionEngine, Normalize, RedTeam, PliniusCommon, GandalfSolutions, AutoTemp, HyperTune, I-LLM, Streaming, Veritas, LeakHub, Claritas, Ouroborous, Config, Lazy, Watcher, Middleware, RateLimiter, I18n, Recovery, Document, Filesystem, TOON, ...).
- dependencies/HelixDevelopment/** — own-org submodules (DocProcessor, LLMOrchestrator, LLMProvider, VisionEngine, LLMsVerifier, Models, HelixMemory, HelixSpecifier, HelixLLM, DebateOrchestrator, ...).

EXCLUDED — third-party vendored submodules (per §11.4.74 no-match → vendor):

- dependencies/LLama_CPP/** — git@github.com:ggml-org/llama.cpp.git
- dependencies/Ollama/** — git@github.com:ollama/ollama.git
- dependencies/HuggingFace_Hub/** — git@github.com:huggingface/huggingface_hub.git

Credential/secret exclusions (per §11.4.10, belt-and-suspenders):

the include list is code-extensions only (no .env / .key / .pem), and the exclude list additionally pins **/.env, **/.env.*, **/*.key, **/*.pem, **/secrets/** so no credential path can ever be indexed.

Event ledger

(events appended below by the automation; newest at the bottom)

2026-05-28T12:03:15Z — HXC-017 config fix: include own-org submodules, exclude only third-party (§11.4.79)

- **Defect:** `.codegraph/config.json` carried a blanket `dependencies/** exclude` that wrongly removed ALL own-org submodules (`dependencies/vasic-digital/** + dependencies/HelixDevelopment/**`) from the index — a §11.4.79 violation (own-org submodules MUST be INCLUDED; only third-party submodules excluded).
- **Fix:** replaced the blanket `dependencies/** exclude` with three specific third-party excludes (`dependencies/LLama_CPP/**`, `dependencies/Ollama/**`, `dependencies/HuggingFace_Hub/**`); added explicit credential excludes (`**/.env`, `**/.env.*`, `**/*.key`, `**/*.pem`, `**/secrets/**`) per §11.4.10.
- **Config JSON validity:** confirmed via `python3 -c "import json; json.load(open('.codegraph/config.json'))"` → VALID.
- **Index status BEFORE re-index:** Files 39,024 / Nodes 624,103 / Edges 1,643,200 / DB 1609.00 MB.
- **Index status AFTER re-index** (`codegraph index .`, exit 0): Files 76,044 / Nodes 1,255,974 / Edges 3,955,444 / DB 2617.24 MB. The +37,020 files / +631,871 nodes delta is the newly-included own-org submodule trees.
- **§11.4.79 anti-bluff probe (own-org symbol now resolves):**
 - `codegraph query EventBus` → `submodules/event_bus/pkg/bus/bus.go:85` (would NOT have resolved under the old blanket `dependencies/** exclude`).
 - `codegraph query helix_memory` → `submodules/helix_memory/pkg/config/config.go (+more)` .
 - `codegraph query llama` filtered to `dependencies/LLama_CPP` → **empty** (third-party correctly excluded).
- **HXC-017 status:** fully done — config fixed, re-index complete, own-org inclusion proven, third-party exclusion proven.

2026-05-29T06:37:00Z — codegraph_sync.sh @ /run/media/milosvasic/DATA4TB/Projects/ HelixCode

FAIL — codegraph sync exited non-zero. Tail of log:\n\n``\n└ Syncing
CodeGraph [2m| [0m (raw progress-spinner log line removed — was
25387 chars of ANSI noise from codegraph_sync.sh; see qa-results/ logs)
(raw progress-spinner log line removed — was 3007584 chars of ANSI
noise from codegraph_sync.sh; see qa-results/ logs)

2026-05-29 — codegraph 0.9.7 update: index/ sync FAIL + §11.4.79 own-org regression (HXC-033)

Event: operator installed codegraph **0.9.7** on the host (codegraph --
version → 0.9.7).

§11.4.80 post-update sync — FAILED (honest ledger, no bluff):

- The 0.9.7 install reset the gitignored index DB (was 76,044 files / 1,255,974 nodes at HXC-017; dropped to 39,203).
- constitution/scripts/codegraph_sync.sh ran codegraph sync . → exited before completing its 4 steps (index reached 43,073 files).
- Full re-index codegraph index . → process **KILLED mid-run, no diagnostic / no exit code** (terminated by signal); index left at 54,207 files. Reproduced.
- codegraph index . --force --quiet → **KILLED again, no diagnostic**; --force wiped + left only 4,630 files.
- codegraph sync . --quiet → **exit 1** at 8,461 files.
- Host memory ample (51 GiB free) — not an obvious §12.6 OOM.

§11.4.79 anti-bluff probe — FAILS: codegraph_search
BundleTranslator (MCP) returns ONLY helix_code/internal/tools/
askuser/... — the own-org submodules/llm_orchestrator/pkg/i18n/
bundle.go symbol does NOT resolve. Own-org submodules are NOT
reachable in the 0.9.7 index. **This is a §11.4.79 regression introduced
by the 0.9.7 update.**

Not a config regression: tracked .codegraph/config.json is intact (git-
clean) — own-org includes + §11.4.10 credential excludes (**/.env, **/
*.key, **/secrets/**) all present.

Root cause: UNCONFIRMED (§11.4.6) — codegraph 0.9.7 index/sync terminate without an actionable diagnostic on this 76k-file repo. Not determinable from captured evidence whether it is a 0.9.7 stability bug, a submodule-traversal change, or a config-schema change. Filed as **HXC-033**; needs operator decision (downgrade to prior working version / upstream bug report / accept degraded index). Evidence: qa-results/codegraph_index_*.log, codegraph_recover_*.log.

2026-05-29 — codegraph 0.9.7 RESOLVED: wipe + init + re-index restored own-org index (HXC-033 → Fixed)

Resolution (operator-directed: "clear all indexed data fully and re-index — it MUST be a data-compatibility problem; ALWAYS index main + HelixDevelopment + vasic-digital"). Confirmed correct.

Root cause (now CONFIRMED, was UNCONFIRMED): codegraph 0.9.7 **requires an explicit codegraph init** before index (behavioral change vs the prior version). The pre-0.9.7 index DB was incompatible; operating on it produced the failures. After a full wipe of the gitignored DB (codegraph.db/-wal/-shm; the 1.7 GB stale WAL was a tell) + codegraph init (tracked config.json preserved — own-org includes + §11.4.10 credential excludes intact) + codegraph index ., the index rebuilt cleanly.

Two earlier mis-diagnoses corrected (per §11.4.6 — stated here as the record): (1) "index crashes/killed mid-run" was a FAULTY pgrep pattern (codegraph index vs the real node ... codegraph.js index .) giving false "ENDED" reads — the process was simply slow (76k files); real ps showed it alive (~600 MB RSS, healthy, climbing). (2) "own-org symbols do not resolve" used the WRONG CLI verb (codegraph search — removed in 0.9.7; the verb is codegraph query) AND queried a stale MCP-server DB handle from before the re-index.

Result (CLI codegraph status on fresh 0.9.7 DB): Files 75,663 / Nodes 1,272,492 / Edges finalizing (1.7M → toward ~3.95M; edge-enrichment phase runs async after node indexing).

§11.4.79 anti-bluff probe — PASS (codegraph query, fresh DB):

- NewBundleTranslator → submodules/llm_orchestrator/pkg/i18n/bundle.go:34 (+ dependencies/vasic-digital/..., doc_processor) — 10 own-org hits. HelixDevelopment + vasic-digital both reachable.

- EventBus → resolves.
- llama_model_load under dependencies/LLama_CPP → empty. third-party correctly excluded.

Status.md hygiene fix: this ledger had bloated to 3.66 MB — a single 3,007,584-char line of raw ANSI progress-spinner output that constitution/scripts/codegraph_sync.sh dumped verbatim on its earlier FAIL. Stripped to 8 KB (all 7 real ledger entries preserved). FOLLOW-UP: codegraph_sync.sh should strip ANSI / not dump raw spinner logs into Status.md (constitution-submodule fix per §11.4.26).

Operational follow-up: the agent-facing codegraph **MCP server** (tools/codegraph/...serve --mcp, a separate install) holds the pre-wipe DB inode — it must be **restarted** to serve the fresh index to AI agents; the CLI query already reflects it.

2026-07-07 — codegraph 1.2.0 Phase-4 reindex + §11.4.79 own-org symbol proof (deferred item completed; HXC-041 opened)

Event: completed the deferred CodeGraph reindex + own-org symbol-resolution proof (RESUME.md #5 — deferred to avoid a stale-v1.1.1-daemon DB conflict). Host now runs codegraph **1.2.0** (matches §11.4.80 update expectation); the serving serve --mcp daemon and the DB backend are both 1.2.0, so the stale-v1.1.1-daemon blocker is resolved.

§11.4.80 reindex/sync — GREEN. codegraph sync synced 8 changed files in 5.6 s (exit 0). Index intact at 1.2.0: **Files 102,657 / Nodes 1,785,100 / Edges 1,918,963 / DB 6.33 GB / node:sqlite full-WAL.**
Evidence: docs/qa/phase4_codegraph_20260707/
{00_pre_sync_status,10_sync_run,11_post_sync_status}.txt.

§11.4.79 own-org symbol resolution — PROVEN (MCP + CLI, unforgeable). Resolved symbols that live ONLY inside own-org submodules:

- admit — an **unexported** Go function — → submodules/helix_llm/internal/vrambroker/broker.go:178 (module github.com/HelixDevelopment/HelixLLM), with verbatim source + blast-radius (callers Acquire, TestAdmit_TruthTable, TestAdmit_PairedMutation).
- ResolveModelCapability → submodules/llms_verifier/llm-verifier/capabilities/registry_resolve.go:62 (module digital.vasic.llmsverifier). Both via the mcp__codegraph__codegraph_explore MCP tool AND the codegraph

query/codegraph node CLI. scripts/codegraph_validate.sh
independently confirms own-org inclusion (helix_qa 28,333 /
llm_provider 151 / constitution 84 / challenges / containers /
security all indexed) — 26 PASS. Evidence: docs/qa/
phase4_codegraph_20260707/
{20_own_org_symbol_proof_cli,21_own_org_symbol_proof_mcp,60_cod
egraph_validate}.txt.

§11.4.10 credentials — CLEAN. Live-DB audit: 0 indexed

.env / .pem / .key files. (The `**/secrets/** + **/.env.*` glob would match some third-party *source* files like .env.d.ts and secrets/ React dirs, but no real credential file types are indexed; the DB is gitignored per §11.4.77, so nothing reaches git.)

§11.4.79 third-party exclusion — PARTIAL FAIL → HXC-041

(BLOCKED on host resources, honest, no bluff). scripts/
codegraph_validate.sh reports 3 FAIL: live index still holds **36,089**
cli_agents + **519** cli_agents_resources + **9** github_pages_website third-
party files. Root cause (FACT, §11.4.102): (1) .codegraph/config.json
exclude is **INERT** in codegraph 1.2.0 — exclusion is .gitignore-driven
per §11.4.78, and these are *tracked* reference dirs not in .gitignore; (2)
the .codegraph/config.json exclude list was recently expanded (git diff:
added tools/opensource/**, submodules/helix_agent/cli_agents/**,
external/**) but the from-scratch codegraph index to apply it was
deferred — codegraph sync is incremental and does not purge now-
excluded files (indexed_at: cli_agents 1783289990159 / Jul-5 vs helix_llm
1783420230150 / Jul-7). **Remediation blocked:** a from-scratch
codegraph index fork-failed (errno=11, runtime: failed to create new
OS thread) — host at **4069/4096** user processes (ulimit -u), saturation
dominated by ~14+ non-ours 75-thread processes that §11.4.174 forbids
killing. The aborted index fork-failed **before writing** — the 1.2.0 sync'd
index is verified INTACT + still resolves own-org symbols (docs/qa/
phase4_codegraph_20260707/50_post_abort_integrity.txt). HXC-041 is
deferred to a low-host-load window; it does NOT affect own-org
reachability (proven above). Evidence: docs/qa/
phase4_codegraph_20260707/
{30_stale_index_rootcause,40_full_index_run,50_post_abort_integrit
y}.txt.